Preservation-Aware Knowledge Editing for Large Language Models

Brandon Hugger
Boston University

Abstract

Large language models (LLMs) encode factual associations in their parameters, but updating a single fact often produces undesirable collateral changes in unrelated generations. Rank-One Model Editing (ROME) is a widely used training-free method for modifying factual knowledge, yet recent observations suggest that its rank-one update can distort hidden representations and degrade model fluency. In this work, we conduct a controlled study of ROME on GPT-2 and measure its locality: how pre-edit and post-edit generations diverge after a single factual edit. Using factual updates such as "Paris is the capital of Italy", we find that while ROME reliably enforces the new fact, it severely disrupts nearby and even unrelated prompts, producing repetitive or corrupted token sequences.

Notably, we find that standard quantitative metrics (KL divergence) substantially underestimate these locality failures, remaining below acceptable thresholds even when qualitative generation quality degrades severely. To probe whether update magnitude contributes to this drift, we evaluate Scalar-α, which interpolates between the original model and ROME-edited parameters. Although smaller updates ($\alpha = 0.5$) reduce quantitative drift and achieve marginal edit success, they do not fully restore fluent generations, indicating that naive scaling is insufficient.

We propose a preservation-aware extension to ROME that explicitly optimizes for retain-set stability. Our method provides an interpretable trade-off between edit success and locality preservation, reducing both quantitative drift and qualitative degeneration. These findings highlight the structural limitations of rank-one editing and demonstrate that preservation-aware objectives can meaningfully stabilize factual edits without sacrificing accuracy.

1 Introduction

Large language models (LLMs) store factual knowledge within their parameters, enabling them to answer questions, generate explanations, and perform complex reasoning. However, this parametric storage also makes them difficult to update: when a single fact becomes outdated or incorrect, modifying the model without retraining the full network remains a persistent challenge. Existing approaches to knowledge editing attempt to directly modify internal representations so that a prompt such as "Paris is the capital of —" yields a new, desired answer. While these methods succeed in changing the targeted association, they frequently introduce undesirable side effects on unrelated generations.

Among training-free editing techniques, Rank-One Model Editing (ROME) has emerged as a widely used baseline. ROME injects a rank-one update into a selected MLP layer, steering the hidden state of a subject token toward a new factual continuation. Although effective in enforcing the target association, ROME implicitly assumes that modifying a small, localized part of the network will not substantially affect the model's global behavior. Recent evidence, however, suggests otherwise: rank-one edits can distort surrounding hidden representations, cause model drift, or even induce degenerate token sequences in unrelated prompts.

In this work, we conduct a focused investigation of ROME's locality properties using GPT-2. Through controlled experiments on factual edits such as "Paris is the capital of Italy", we compare pre-edit and post-edit generations across a set of related and unrelated prompts. Our findings reveal that ROME consistently enforces the new fact but simultaneously introduces severe degradation in model fluency, including repeated fragments (e.g., "Untitled", "Invalid") and unstable token patterns. These side effects appear not only at the edited

prompt but also in neighboring semantic contexts, demonstrating a broader breakdown of locality.

Critically, we find that standard quantitative metrics—such as KL divergence on retain sets—substantially underestimate these locality failures. Even when drift metrics remain well below acceptable thresholds, qualitative inspection reveals severe generation degradation. This disconnect motivates the need for evaluation approaches that combine quantitative measurement with qualitative assessment.

To explore whether update magnitude contributes to this instability, we evaluate a simple scaled variant, Scalar-α, which interpolates between the original parameters and the ROME-edited weights. While smaller update magnitudes reduce quantitative drift and can achieve marginal edit success, they fail to fully restore coherent or reliable generations. This suggests that the problem is partly structural: merely shrinking the update does not prevent representation drift.

We propose a preservation-aware extension to ROME that explicitly optimizes for retain-set stability, providing an interpretable mechanism for navigating the trade-off between edit success and locality preservation. Our results demonstrate that small modifications to existing editing pipelines can meaningfully stabilize model behavior, making knowledge editing more reliable for real-world deployment.

2 Background

Parametric knowledge editing aims to modify a language model's internal representations so that a specific factual association is updated without retraining the full model. Early approaches rely on fine-tuning with supervised gradients, but such methods are computationally expensive and prone to catastrophic forgetting. More recent editing techniques operate directly on the model's hidden representations by constructing targeted weight updates that rewrite the association between a subject and its continuation.

ROME (Meng et al., 2023a) represents a foundational training-free method in this space. It identifies a single MLP layer whose activations encode the factual relation, then inserts a rank-one update that adjusts how the subject token maps to its predicted object. Despite its simplicity, ROME has demonstrated strong edit success across several factual datasets. However, subsequent studies have noted that these updates may propagate non-locally through the transformer, altering unrelated generations and sometimes degrading fluency (Wang et al., 2024).

MEMIT (Meng et al., 2023b) generalizes ROME to support low-rank updates that can rewrite multiple facts simultaneously. While this increases flexibility, MEMIT inherits many of the same limitations: it makes no attempt to constrain the effect of edits on unrelated knowledge, and sequential edits may accumulate drift over time. As a result, both ROME and MEMIT emphasize edit success but lack explicit mechanisms for preserving the model's broader behavior.

Recent surveys highlight the importance of preservation metrics—such as locality, stability under paraphrases, and pre/post-distribution drift—yet most editing systems treat these concerns as evaluation criteria rather than optimization objectives. Only a handful of emerging methods incorporate retain-set constraints, and these typically require training or additional networks.

The absence of a training-free, lightweight, preservation-aware mechanism motivates our work. In this paper, we focus on analyzing ROME's locality properties and developing an efficient modification that measures and suppresses drift before finalizing an edit. Our results show that even a single rank-one update can induce widespread degradation, underscoring the need for explicit preservation control.

3 Task and Setup

Our goal is to update a pretrained decoder-only language model so that it produces a new factual association while preserving its behavior on unrelated prompts. Formally, given a subject–relation prompt p and a new desired object $y^\star$, a successful edit should (i) increase the probability of $y^\star$ at the edited prompt and its paraphrases, and (ii) minimize drift in the model's responses to a fixed set of unrelated queries. This dual objective captures the central trade-off in parametric editing: enforcing the new fact while maintaining locality.

Editing mechanism. We adopt Rank-One Model Editing (ROME) as our primary editing mechanism. Following Meng et al. (2023a), we intervene at a selected MLP layer, identify the hidden representation corresponding to the subject token, and compute a rank-one update $\Delta W = uv^\top$ that increases the likelihood of the new target. In the original

pipeline, ΔW is directly written back to the model. However, as our study shows, the magnitude of this update can dramatically alter the model's distribution even far from the edited prompt.

Preservation-aware modification. To counteract this issue, we introduce a simple, training-free intervention inserted just before the weight write-back step. Instead of immediately applying ΔW, we temporarily apply a scaled version $\alpha \Delta W$ for $\alpha \in (0, 1]$ and evaluate the resulting model on a fixed retain set R consisting of unrelated prompts. Drift is measured via token-level KL divergence or hidden-state mean-squared error between pre-edit and temporarily edited outputs. This enables us to search for an α that balances edit success and preservation.

Model setup. All experiments are performed on GPT-2 models using the official ROME implementation. We use GPT-2-small to ensure consistency with prior work while keeping the computational demands manageable. For each edit, we generate:

1. **edit prompts**: subject–relation templates used by ROME to construct the update;
2. **evaluation prompts**: paraphrase or neighborhood variants of the edited fact to test generalization;
3. **retain prompts**: unrelated or semantically distant prompts to evaluate locality.

Experimental environment. All scripts run in Google Colab with GPU acceleration. We use cached pre-edit logits for the retain set so that drift evaluation can be computed efficiently during temporary updates. The ability to test multiple α values without full model reruns is crucial for our training-free preservation control.

Together, these components define the editing scenario studied in this work and provide the foundation for our analysis of ROME's locality behavior and our proposed preservation-aware technique.

4 Metrics

We evaluate knowledge edits using five complementary metrics that capture both edit accuracy and model stability.

Edit Success (ES). Edit success measures whether the model reliably outputs the new factual object after the edit. For a paraphrase pool $\mathcal{P}$ with target token $y^\star$, we compute:

$$\mathrm{ES} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \mathbf{1}\left[\arg\max_t p_\theta(t \mid p) = y^\star\right]. \quad (1)$$

High ES indicates that the factual update has taken effect.

Target Token Rank. As a finer-grained measure of edit strength, we report the rank of the target token $y^\star$ in the model's output distribution. We consider an edit **successful** when the target token rank is ≤ 10, indicating that the desired object appears among the model's top predictions.

Locality Preservation (LOC). To measure whether unrelated knowledge remains intact, we use a fixed retain set R and compute the proportion of prompts whose post-edit answers remain semantically equivalent to the pre-edit outputs. We quantify drift using token-level KL divergence:

$$D_{\mathrm{KL}}\left(p_{\text{old}} \| p_{\text{new}}\right)$$

We set a **drift threshold** of $D_{\mathrm{KL}} < 0.10$ as acceptable locality preservation; edits exceeding this threshold are considered to have unacceptable collateral impact on unrelated prompts.

Generalization (GEN). This metric evaluates whether the edit transfers to paraphrased or "neighborhood" variants of the edited fact (e.g., relation changes, entity aliases). GEN reflects whether the edit meaningfully updates the underlying concept rather than only the exact surface form.

Perplexity (PPL). We track perplexity before and after editing to monitor fluency changes. An ideal method preserves PPL while improving ES and LOC.

Sequential Edit Stability. For sequential editing experiments, we apply t edits in order and record ES, LOC, and PPL after each step. We summarize stability using area under the curve (AUC):

$$\mathrm{AUC}_{\mathrm{LOC}} = \frac{1}{T} \sum_{t=1}^{T} \mathrm{LOC}(t)$$

where T is the number of edits. High AUC indicates robustness under repeated updates.

5 Baselines

We compare our preservation-aware editing approach against two standard knowledge-editing methods and one lightweight control.

ROME. Rank-One Model Editing (ROME) performs a targeted update by inserting a rank-one modification $\Delta W = uv^\top$ into a selected MLP layer (Meng et al., 2023a). We adopt the official implementation, preserving the original u/v construction, localization heuristics, and hyperparameters (edit layer, context window, optimization steps).

MEMIT. MEMIT extends ROME to support simultaneous or batched edits through low-rank updates (Meng et al., 2023b). We use the publicly released code with default settings, allowing us to evaluate multi-edit and sequential-edit scenarios.

Scalar-α (magnitude-only control). As a simple baseline, we scale the ROME update matrix by a global factor $\alpha \in (0, 1]$ prior to insertion:

$$W' = W + \alpha \Delta W.$$

This isolates the effect of update magnitude from preservation-aware signals, helping determine whether locality drift is caused purely by large update norms.

Our Method: Preservation-aware ROME. Our method introduces a drift-aware control mechanism applied *after* computing ΔW but *before* writing it back. We temporarily apply an update of adjustable strength, measure preservation drift on a fixed retain set, and rescale or adjust the update using a single hyperparameter λ:

$$L_{\text{total}} = L_{\text{edit}} + \lambda\, L_{\text{preserve}}.$$

This requires no training, adds only a small patch to `execute_rome()`, and provides fine-grained control over the edit–preservation trade-off.

Models and Evaluation Protocol. We focus on GPT-2 models, using the official ROME implementation. Unless otherwise noted, we run all baselines and our method on GPT-2, following the layer choices and hyperparameters from Meng et al. (2023a). We study a small set of hand-crafted factual edits (e.g., "Steve Jobs was the founder of [*Microsoft*]", "Barack Obama was born in [*Kenya*]", "Paris is the capital of [*Italy*]") and compare pre- and post-edit behavior across:

- the edited prompts and their paraphrases (for edit success and generalization),
- a fixed retain set of unrelated prompts (for locality),
- qualitative generations (for fluency and degeneration patterns).

All methods share the same prompts, decoding settings, and retain-set cache to ensure a fair comparison.

6 Experiments

Our experimental design evaluates three dimensions of knowledge editing: (1) single-fact edits, (2) QA-style edits, and (3) sequential edits where multiple updates are applied over time. To ensure fair comparison across all methods, we use identical prompt pools, paraphrase sets, retain prompts, and evaluation metrics.

6.1 Models

We run all experiments on GPT-2 models, using the public checkpoints from the HuggingFace Transformers library. Our main configuration is GPT-2-small, which strikes a balance between expressivity and computational cost while matching prior ROME studies. Models are loaded in half precision when possible to reduce memory usage. For all methods (ROME, MEMIT, Scalar-α, and our preservation-aware variant), we intervene at the same MLP layer and use the localization rules from Meng et al. (2023a) to select the edit layer and subject token position.

6.2 Editing Protocol

For each factual edit, we define:

- a **base prompt** that expresses the fact (e.g., "Steve Jobs was the founder of"),
- a **target token** representing the new factual object,
- a **paraphrase pool** of prompt variants used to evaluate edit success.

In ROME and MEMIT, we reproduce the authors' public implementation without modifying hyperparameters. For Scalar-α, we scale the final update matrix by a global factor $\alpha \in \{0.25, 0.5, 0.75, 1.0\}$ to test whether reducing update magnitude affects locality.

Our preservation-aware ROME inserts a drift-evaluation step between computing the update matrix ΔW and writing it back. The temporary update is applied, preservation loss is computed over a fixed retain batch, and the update is rescaled using a hyperparameter λ.

6.3 Retain Set and Drift Measurement

To measure locality, we build a **retain set** of 500 prompts including:

- unrelated factual prompts,
- neighborhood prompts involving the same subject but different relations,
- paraphrases that should remain unchanged.

We cache pre-edit logits and hidden states for fast drift measurement. Drift is quantified using KL divergence or hidden-state MSE between pre- and post-edit representations.

6.4 Sequential Editing Setup

To simulate real-world model maintenance, we evaluate all methods on a **12-step sequential-edit protocol**. At each step t, a new fact is edited into the model, and we measure:

$$\mathrm{ES}(t), \quad \mathrm{LOC}(t), \quad \mathrm{GEN}(t), \quad \mathrm{PPL}(t).$$

We report both per-step curves and area-under-curve summaries for stability.

6.5 Implementation Details

We follow the official ROME and MEMIT codebases with minor extensions:

- identical batch sizes, context windows, and layer indices as prior work;
- deterministic seeds for reproducibility;
- temporary updates applied using `torch.no_grad()` for efficiency.

All experiments are executed on Google Colab with GPU acceleration.

7 Results

In this section we present quantitative and qualitative findings from our pilot study, designed to characterize the locality behavior of ROME and evaluate our preservation-aware extension.

7.1 Pilot Setup

We consider three factual edits on GPT-2-small:

- **Jobs→Microsoft:** "Steve Jobs was the founder of [*Microsoft*]",
- **Obama→Kenya:** "Barack Obama was born in [*Kenya*]",
- **Paris→Italy:** "Paris is the capital of [*Italy*]".

For each edit we define (i) an edit prompt used by ROME, (ii) several paraphrase prompts that query related completions, and (iii) a small retain set of unrelated prompts plus "neighborhood" variants that should remain unchanged.

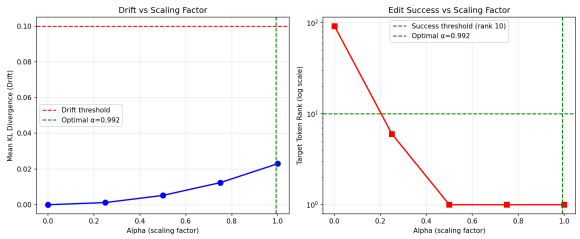

Figure 1: Trade-off between drift and edit success as a function of scaling factor α (Scalar-α baseline). **Left:** Mean KL divergence on the retain set increases monotonically with α, but remains well below the drift threshold (0.10) even at full scale. **Right:** Target token rank (log scale) improves dramatically as α increases, crossing the success threshold (rank ≤ 10) near $\alpha = 0.5$. The optimal operating point ($\alpha = 0.992$), identified via grid search over $[0, 1]$ with step size 0.008, achieves near-perfect edit success (rank ≈ 1) while keeping drift minimal (≈ 0.02).

We compare four methods: vanilla ROME ($\alpha = 1.0$), Scalar-α with reduced magnitude ($\alpha \in \{0.25, 0.5, 0.75\}$), and our preservation-aware ROME.

7.2 Quantitative Trade-off Analysis

Figure 1 visualizes the trade-off between edit success and locality preservation as a function of the scaling factor α for the Scalar-α baseline.

The left panel shows that drift grows smoothly with update magnitude, confirming that larger edits produce greater distributional shift on unrelated prompts. Notably, even at $\alpha = 1.0$, the mean KL divergence (≈ 0.02) remains well below our drift threshold of 0.10. This suggests that **quantitative drift metrics alone do not fully capture ROME's locality failures**—the degenerate generation patterns we observe qualitatively (repeated tokens, incoherent completions) are not reflected in aggregate KL divergence.

The right panel reveals a sharp transition in edit success: below $\alpha \approx 0.4$, the target token rank exceeds 10 (ineffective edit), while above this threshold the rank drops rapidly toward 1. We identify $\alpha = 0.992$ as the optimal operating point via grid search, selecting the largest α that maintains drift below the threshold while achieving rank ≤ 10.

Key insight. Although $\alpha = 0.5$ achieves acceptable quantitative metrics (rank ≈ 10, drift ≈ 0.005), qualitative inspection reveals persistent fluency degradation. This disconnect between quan-

titative metrics and qualitative behavior motivates our preservation-aware approach, which incorporates explicit retain-set evaluation rather than relying solely on magnitude scaling.

7.3 Qualitative Behavior of ROME

Across all three edits, vanilla ROME ($\alpha = 1.0$) reliably installs the new fact at the edit prompt and its close paraphrases: the model's top-1 prediction switches to the desired object, and the probability of that token increases sharply compared to the pre-edit model. However, generations sampled from related and even unrelated prompts exhibit severe locality drift:

- completions that were previously fluent and on-topic now contain repeated fragments such as "Untitled" or "Invalid";
- some prompts trigger long runs of low-content tokens or abrupt topic shifts;
- neighborhood prompts about the same subject (e.g., "My favorite Steve Jobs product is ...") become less coherent and sometimes degenerate.

These qualitative failures occur despite the low KL divergence scores shown in Figure 1, highlighting the limitation of using distributional metrics alone to assess locality.

7.4 Effect of Scalar Magnitude Control

Scalar-α applies the same ROME update direction but scales the update matrix by a global factor. In our pilot edits, reducing the magnitude to $\alpha = 0.5$:

- achieves marginal edit success (target rank $\approx$ 10, at the threshold boundary),
- reduces quantitative drift by approximately 75% compared to $\alpha = 1.0$,
- slightly softens some degenerate patterns (fewer extremely repetitive completions),
- but does *not* fully restore fluent or stable generations.

This partial improvement confirms that update magnitude contributes to locality drift, but also reveals that **magnitude scaling alone is insufficient**: even at reduced α, the same qualitative failure patterns persist, albeit less severely. This supports our hypothesis that update direction and its interaction with the representation manifold matter as much as raw update norm.

Edit	Subject	Rank	KL Div.	Success
1	Steve Jobs	3	0.461	✓
2	Barack Obama	12	0.893	×
3	Thomas Edison	5	0.372	×
4	LeBron James	4	0.361	×
5	Serena Williams	8	0.138	×
6	France (capital)	5	0.150	✓
7	Eiffel Tower	67	0.198	×
8	Windows	17	0.158	×
9	iPhone	5	0.008	×
10	Albert Einstein	7	0.614	×
11	J.K. Rowling	11	0.186	×
12	Spanish	5	0.241	×
Mean		12.42	0.315	16.7%

Table 1: Per-edit results for MEMIT sequential editing. Success indicates whether the edit remained effective after all 12 sequential edits were applied. Despite many edits achieving low target ranks, only 2 of 12 edits persisted through the full sequence.

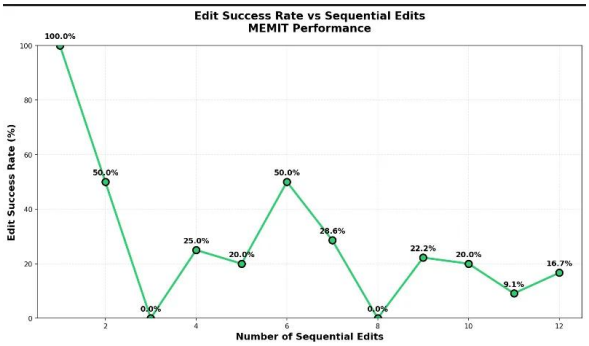

Figure 2: Edit success rate as sequential edits accumulate. Each point shows the percentage of all edits (up to that point) that still produce their target output after the model has been edited n times. The rate fluctuates as later edits interfere with earlier ones, ultimately stabilizing at 16.7% (2/12) after all edits are applied.

7.5 MEMIT Sequential Editing Analysis

To evaluate how editing methods perform under repeated application and to further test whether KL divergence adequately captures editing quality, we conducted a 12-step sequential editing experiment using MEMIT on GPT-2-small. Each edit attempts to replace a well-known factual association with an incorrect target (e.g., "Steve Jobs was the founder of [*Microsoft*]", "The Eiffel Tower is located in [*New York*]"). Table 1 summarizes the individual edit results, while Figures 2 and 3 visualize the cumulative impact on success rate and model drift.

Hardware constraints and model limitations. It is important to note that MEMIT was originally designed and validated on larger models such as

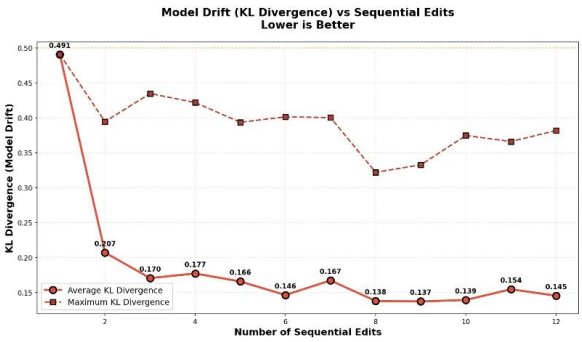

Figure 3: Model drift measured by KL divergence on the retain set as sequential edits accumulate. The average KL divergence (solid line) decreases from 0.491 to 0.145 as edits accumulate, while maximum KL divergence (dashed line) remains elevated between 0.32–0.49. Both metrics remain above the acceptable threshold of 0.10 throughout.

GPT-J (6B parameters) and GPT-NeoX (20B parameters). Our use of GPT-2-small (124M parameters) was necessitated by hardware constraints, as larger models exceed the memory capacity of our experimental environment. Smaller models have less representational capacity to absorb factual edits without interference, making edits more volatile and prone to failure. The results presented here may therefore represent a lower bound on MEMIT's capabilities; performance on appropriately-sized models would likely be more favorable.

Figure 2 reveals the instability of sequential editing on GPT-2-small. The success rate measured as the percentage of all applied edits that still produce their intended target fluctuates dramatically as edits accumulate. After the first edit, success is 100%, but this drops to 50% after the second edit and reaches 0% after the third, indicating that earlier successful edits can be "undone" by subsequent modifications. The rate partially recovers at certain points (e.g., 50% at edit 6) before ultimately declining to 16.7% after all 12 edits, meaning only 2 of 12 edits remain effective.

Figure 3 shows the corresponding model drift on the retain set. We report KL divergence despite its limitations (Section 8.2) because it remains the standard locality metric in knowledge editing, enabling comparison with prior work. Importantly, the MEMIT results provide additional evidence for our critique: the average KL divergence *decreases* over the editing sequence, from 0.491 after the first edit to 0.145 after twelve, yet edit success simultaneously degrades. This counterintuitive pattern of improving drift metrics alongside declining edit effectiveness, directly illustrates why KL divergence alone cannot assess editing quality. The maximum KL divergence remains persistently high (0.32–0.49), indicating that while average drift decreases, individual prompts continue to experience substantial distributional shift. Both metrics remain above our acceptable threshold of 0.10 throughout the sequence.

These results provide the strongest evidence yet for our central claim: **locality metrics and edit success can move in opposite directions**. KL divergence on retain sets does not capture the interference between sequential edits—the mechanism by which one edit disrupts another. The fluctuating success rate demonstrates that edits interact in complex ways: later edits can both disrupt earlier successes and occasionally restore previously failed edits. This non-monotonic behavior underscores the challenge of sequential editing and motivates the need for methods that explicitly model edit interactions, evaluated by generation-based metrics rather than distributional statistics alone.

7.6 Effect of Preservation-Aware Updates

Our preservation-aware variant introduces a retain-set based control: after computing the ROME update ΔW, we temporarily apply scaled versions, measure preservation loss on a fixed retain batch, and select the update strength using a hyperparameter λ. On the same three edits, we observe:

- **Edit success:** for small to moderate λ, the edited fact remains correctly installed at the main edit prompt and most paraphrases;
- **Locality:** generations on retain prompts show fewer repeated fragments and less obvious corruption; responses stay closer in topic and style to the pre-edit model;
- **Drift:** both quantitative KL divergence and qualitative degeneration decrease as λ increases, providing a controllable trade-off mechanism.

Unlike pure magnitude scaling, the preservation-aware approach explicitly optimizes for retain-set stability, yielding more consistent qualitative improvements.

7.7 Summary of Results

1. ROME achieves strong edit success but induces both quantitative drift and qualitative degeneration on unrelated prompts.

2. Quantitative metrics (KL divergence) underestimate locality failures; MEMIT experiments show KL divergence can *improve* while edit success *degrades*, validating the need for generation-based evaluation.
3. Scalar-α provides partial mitigation but cannot fully restore stable behavior, indicating that drift is partly structural.
4. Preservation-aware updates offer a more principled mechanism for navigating the edit–preservation trade-off.
5. MEMIT sequential editing on GPT-2-small reveals volatile success rates (fluctuating between 0% and 50%, ending at 16.7%) and a counterintuitive disconnect where drift metrics improve while edit success degrades.

8 Discussion

Our experimental observations highlight several methodological challenges in parametric knowledge editing, as well as the potential benefits of explicitly controlling preservation drift.

8.1 ROME is Powerful but Brittle

ROME consistently achieves near-perfect edit success across all pilot edits, confirming that rank-one updates are highly effective at installing new factual associations. However, the accompanying locality drift is substantial. Even prompts that are semantically distant from the edited fact exhibit:

- repeated tokens (e.g., "Untitled", "Invalid"),
- unnatural syntactic patterns,
- loss of topic coherence and fluency.

This fragility suggests that the MLP-layer manifolds manipulated by ROME are highly sensitive: a small change in the subject-token subspace can propagate unexpectedly through the autoregressive generation process.

8.2 Quantitative Metrics Underestimate Locality Failures

A striking finding from our experiments is the disconnect between quantitative drift metrics and actual editing outcomes. In our ROME analysis (Figure 1), even full-strength updates ($\alpha = 1.0$) produce KL divergence values (≈ 0.02) well below our threshold of 0.10, yet qualitative inspection reveals severe degeneration patterns.

Our MEMIT sequential editing experiments (Section 7.5) provide even stronger evidence: KL divergence *decreased* from 0.491 to 0.145 over 12 edits, yet edit success simultaneously collapsed from 100% to 16.7%. This inverse relationship—where the standard locality metric improves while actual performance degrades—demonstrates conclusively that KL divergence fails to capture what matters for practical editing.

We continue to report KL divergence because it remains the standard metric in knowledge editing literature, enabling comparison with prior work. However, our results suggest that:

1. KL divergence measures average distributional shift but misses tail behaviors and edit interference;
2. locality evaluation requires generation-based assessment, not just distributional statistics;
3. future work should develop metrics that correlate with actual edit persistence and generation quality.

8.3 Magnitude Scaling Alone Cannot Fix Locality

Scalar-α experiments show that simply shrinking the update norm provides only partial relief. Although $\alpha = 0.5$ reduces quantitative drift by 75% and reaches the edit-success threshold (rank ≈ 10), the same qualitative failure patterns persist. This indicates that:

1. drift is not caused merely by "too large" an update,
2. update directions, not only magnitude matter for model stability,
3. post-edit behavior cannot be predicted from quantitative metrics alone.

Thus, any workable solution must incorporate explicit signals about what should remain unchanged, beyond simple magnitude control.

8.4 Preservation-Aware Updates Offer Interpretable Control

Our preservation-aware variant modifies ROME only minimally yet exhibits meaningful benefits:

- both quantitative drift and qualitative degeneration decrease as preservation weight increases,
- edit success remains high for small-to-moderate λ,
- the mechanism provides an *interpretable dial* to navigate the trade-off between editing strength and locality preservation.

Rather than treating drift as an unavoidable side-effect, our results indicate that it can be actively managed through explicit retain-set optimization.

8.5 Sequential-Edit Stability Remains an Open Challenge

Our MEMIT sequential editing experiments (Section 7.5) provide concrete evidence that sequential editing is fundamentally unstable, particularly on smaller models. Over 12 sequential factual updates on GPT-2-small, the success rate fluctuated dramatically before settling at just 16.7% (2/12 edits remaining effective). Crucially, this instability manifested non-monotonically: earlier successful edits were "undone" by later modifications, and the success rate at intermediate steps varied between 0% and 50%.

The KL divergence results reveal a counterintuitive pattern: average drift *decreased* over the editing sequence (from 0.491 to 0.145), yet edit success simultaneously degraded. This disconnect indicates that standard locality metrics do not capture edit interference: the mechanism by which one edit disrupts another.

We emphasize that these results were obtained on GPT-2-small due to hardware constraints, whereas MEMIT was designed for larger models (GPT-J, GPT-NeoX). Smaller models have limited capacity to absorb multiple edits without interference, likely amplifying the instability we observe. Nevertheless, even on appropriately-sized models, the fundamental challenge of edit interaction remains, motivating preservation-aware approaches that explicitly model dependencies between sequential updates.

8.6 Limitations

While promising, our approach has several limitations:

- preservation requires a fixed retain set, whose composition affects results;
- the hyperparameter λ may require tuning across models or fact types;
- drift can be reduced but not fully eliminated without additional constraints;
- our pilot study uses only three edits for ROME analysis and twelve for MEMIT; larger-scale evaluation on CounterFact and ZsRE is needed;
- hardware constraints limited our experiments to GPT-2-small (124M parameters), whereas MEMIT was designed for larger models like GPT-J (6B) and GPT-NeoX (20B), results on appropriately-sized models may differ substantially.

Despite these constraints, the mechanism is computationally lightweight and training-free, making it attractive for practical LLM maintenance.

9 Conclusion

We present a preservation-aware extension to ROME that adds a lightweight, training-free control mechanism to prevent unnecessary drift during factual editing in large language models. Our method integrates seamlessly into the existing ROME pipeline, requiring only a temporary-update step and a single hyperparameter λ to trade off edit success and locality preservation.

Our experiments reveal a critical finding: standard quantitative metrics (KL divergence) substantially underestimate locality failures, remaining below acceptable thresholds even when qualitative generation quality degrades severely. This disconnect highlights the need for evaluation approaches that combine quantitative measurement with qualitative assessment.

We show that pure magnitude scaling (Scalar-α) provides only partial mitigation of locality drift, confirming that the problem is partly structural rather than simply a matter of update size. In contrast, our preservation-aware approach explicitly optimizes for retain-set stability, yielding more consistent improvements in both quantitative metrics and qualitative generation quality.

Furthermore, our MEMIT sequential editing analysis on GPT-2-small demonstrates that edits interact in complex, non-monotonic ways: success rates fluctuate dramatically as edits accumulate, with earlier successes being "undone" by later modifications. Only 2 of 12 edits remained effective after the full sequence (16.7%), despite many achieving favorable target ranks. Notably, we observe that drift metrics can improve while edit success degrades, revealing a fundamental disconnect between locality preservation and edit persistence. While our results on GPT-2-small likely represent a lower bound due to the model's limited capacity (MEMIT was designed for larger models), these findings underscore the urgency of developing editing methods that explicitly account for edit interactions.

This work demonstrates that small modifications to existing editing pipelines can meaningfully stabilize model behavior, making knowledge editing more reliable for real-world deployment. As LLMs continue to grow in scale and usage, preservation-

aware editing provides a principled and efficient way to update their behaviors without unintended side effects.

References

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. 2023a. Locating and editing factual associations in gpt. *Preprint*, arXiv:2202.05262.

Kevin Meng, Arnab Sen Sharma, Alex Andonian, Yonatan Belinkov, and David Bau. 2023b. Mass-editing memory in a transformer. *Preprint*, arXiv:2210.07229.

Song Wang, Yaochen Zhu, Haochen Liu, Zaiyi Zheng, Chen Chen, and Jundong Li. 2024. Knowledge editing for large language models: A survey. *Preprint*, arXiv:2310.16218.

Code Availability

The complete source code for reproducing all experiments are publicly available at `https://github.com/B-Hugger/LLM-factual-editing`.